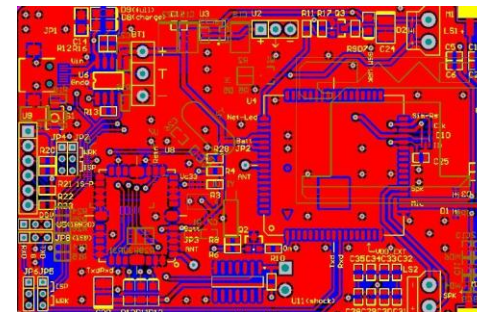
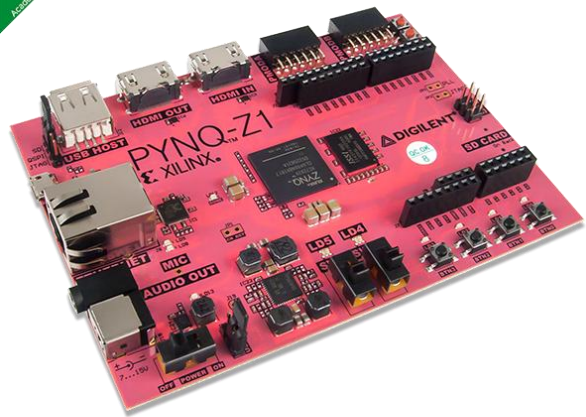
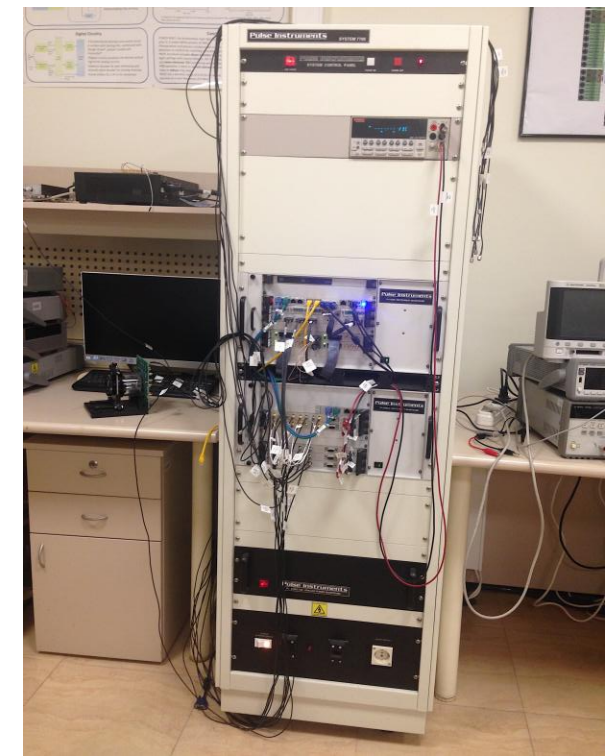




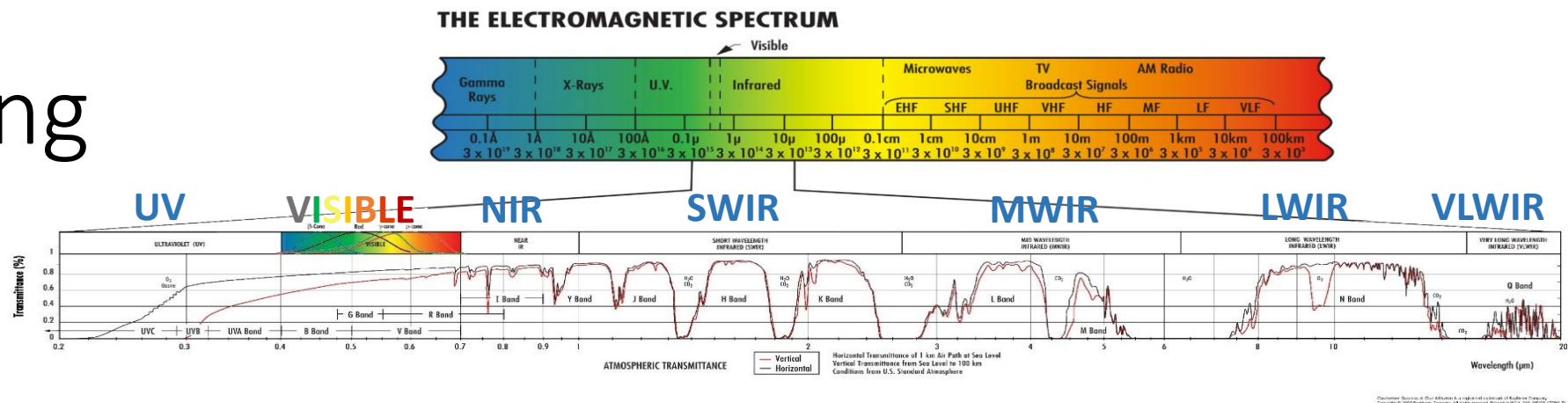
Development of Camera Module for Microbolometer Based Infrared Imaging Applications

Supervisors: Yaşar Gürbüz Ömer Ceylan Melik Yazıcı

Students: Kerem Şahin Muhtasham Oblokoulov Jiyan Baran Bükün



IR Imaging



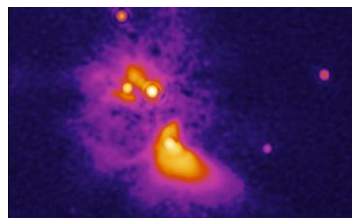
Civilian



automotive



medical



astronomy

Military



surveillance



missile



space

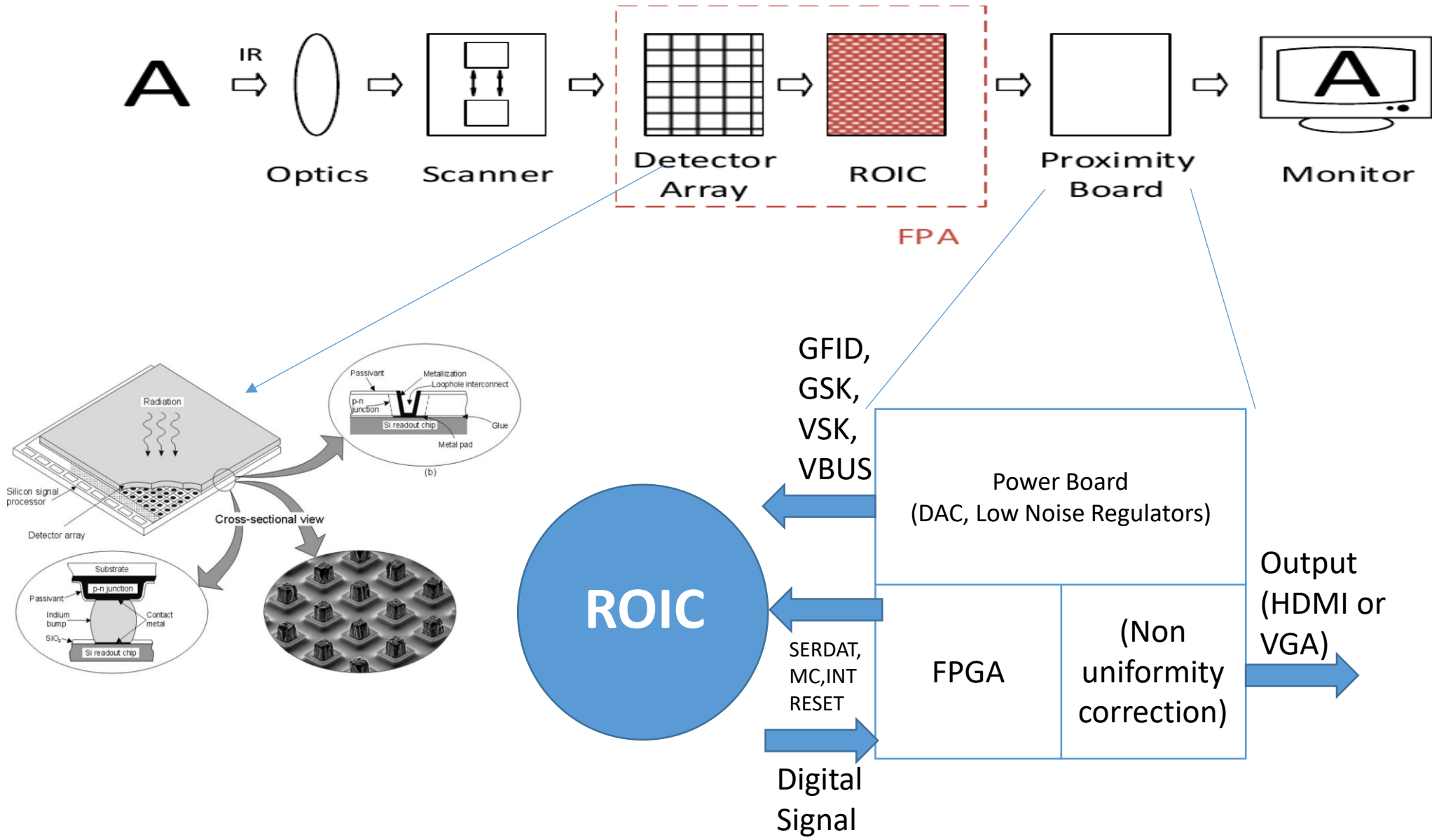


uav

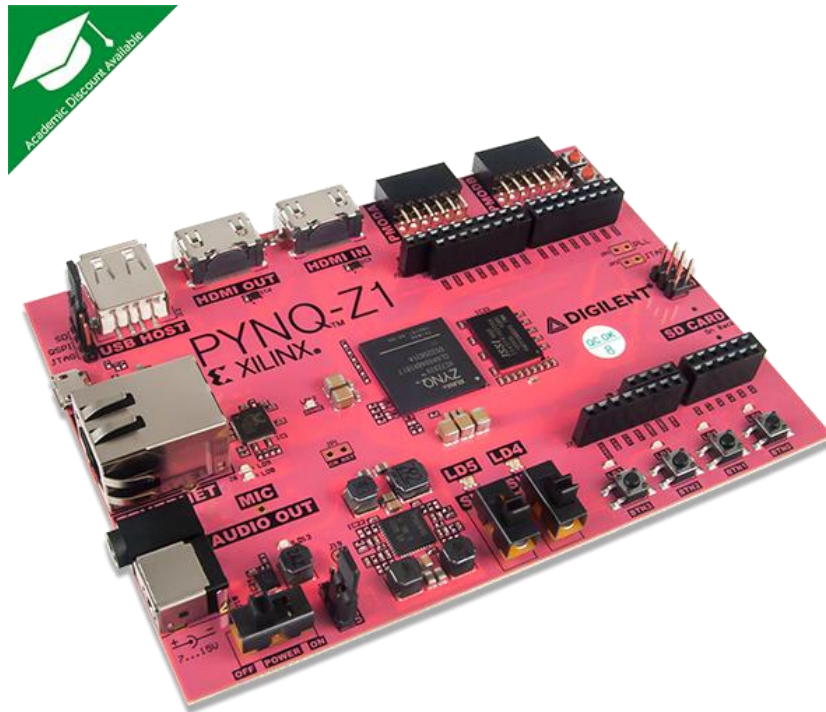


flir

Our System



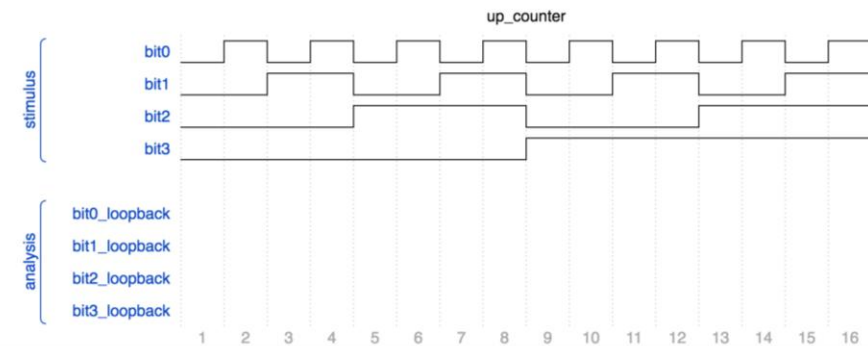
What we did ?



```
In [2]: from pynq.lib.logictools import Waveform
```

```
up_counter = {'signal': [
    ['stimulus',
     {'name': 'bit0', 'pin': 'D0', 'wave': '1h' * 8},
     {'name': 'bit1', 'pin': 'D1', 'wave': '1.h.' * 4},
     {'name': 'bit2', 'pin': 'D2', 'wave': '1...h...' * 2},
     {'name': 'bit3', 'pin': 'D3', 'wave': '1.....h.....'}],
    ['analysis',
     {'name': 'bit0_loopback', 'pin': 'D0'},
     {'name': 'bit1_loopback', 'pin': 'D1'},
     {'name': 'bit2_loopback', 'pin': 'D2'},
     {'name': 'bit3_loopback', 'pin': 'D3'}],
    'foot': {'tock': 1},
    'head': {'text': 'up_counter'}]}

waveform = Waveform(up_counter)
waveform.display()
```



Note: Since there are no captured samples at this moment, the analysis group will be empty.

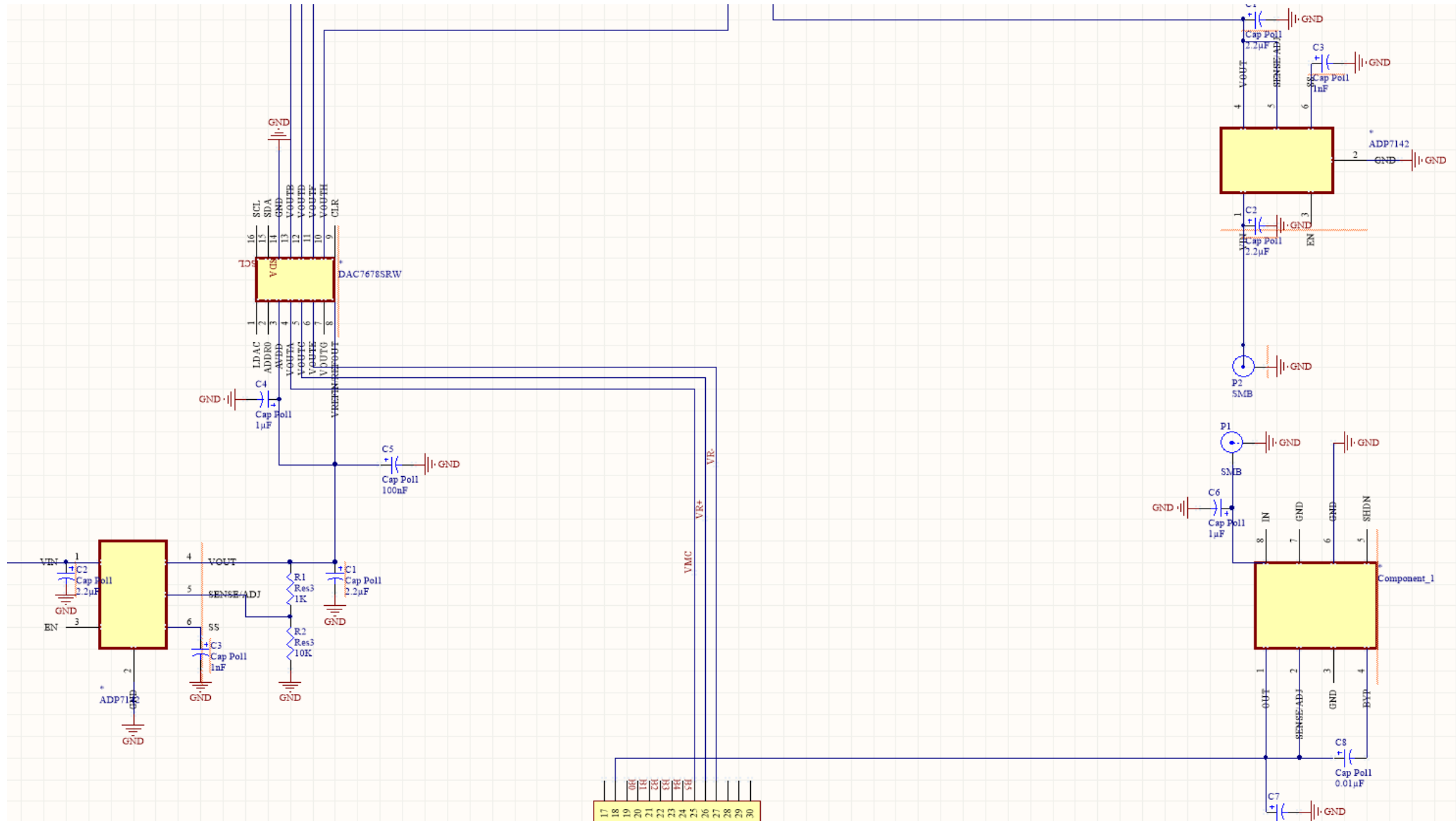
- FPGA versions searched ,optimum version is selected for desired application and started examining sample code for needs

- Components for PCB are selected and supplied

	A	B	C	D	E	F	G	H	I	J
1	Part Number	Used With	Device	Capacitance	Necessity	Material	Type	Price	size	
2	LMK212AB7225MD-T-ND	ADP7142	CAPACITOR	2.2uF	Compulsory	Ceramic	X5R	0.05		
3	GRM033R71E102KA01D	ADP7142	CAPACITOR	0.1uF	Compulsory	Ceramic	X7R	0.1	805	
4										
5										
6	CL03A104KQ3NNNC	DAC7678SPW	CAPACITOR	100nF	Compulsory	Ceramic	X5R	0.1	805	
7	CL10B105KA8NNNC	DAC7678SPW	CAPACITOR	1uF	Compulsory	Ceramic	X7R	0.4	805	
8	UWX1C101MCL1GB	DAC7678SPW	CAPACITOR	100uF	Not Comp	Electrolytic	AECQ200	0.3		
9										
10	cl21 var	LT1763CS8-3.3	CAPACITOR	10uF	Compulsory	Ceramic	X5R			
11	C0402C103K4RACTU	LT1763CS8-3.3	CAPACITOR	0.01uF	Compulsory	Ceramic	X7R			
12		LT1763CS8-3.3	CAPACITOR	1uF	Compulsory	Ceramic	X7R			
13										
14										
15										
16										



- PCB Schematic were drawn



- Verilog code for generating clock signals was written

```
module vco_cos(va, vb);
    input va;
    output vb;
    electrical va, vb;

    parameter real vdc = 1;
    parameter real vac = 0;
    parameter real center_freq = 1K;
    parameter real vco_gain = 1;
    parameter real phase = 0;
    parameter integer step = 32; //Time steps per period

    real frequency;
    real inst_phase;
    real ini_phase;

    analog begin
        frequency = center_freq + V(va) * vco_gain;
        ini_phase = phase * `M_PI / 180.0;
        //bound the time step to be 1/step of the period
        $bound_step(1.0 / (step*frequency));

        inst_phase = idtmod(frequency, 0.0, 1.0, -0.5);
        V(vb) <+ vdc + vac*vac/2.0/vdc*cos(2* `M_PI * inst_phase + ini_phase);
    end
endmodule
```

Plans for next two weeks

- CAM → RAM Coding on PYNQ
- RAM → HDMI OUT Coding on PYNQ
- Drawing PCB layout as planned in schematic
- **FPGA and PCB Integration**
- **Testing And Verification**